# FLY RANK ORGANIZATION

# MACHINE LEARNING

# JULY-INTERN

**OBJECTIVE:**

**THE PROMPT LADDER**

The Prompt Ladder

Topic: explaining a credit-card customer dataset (9,000 customers, 18 behavioral features) before clustering it.

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## Baseline (Version 0)

**Prompt:** "Explain this data."

**Output (what you'd actually get):**

> I'd be happy to help explain the data! However, I don't see any data attached to your message. Could you share the dataset or describe what it contains?

**Notes:**

- What changed in the prompt: nothing yet — this is the starting point.

- What actually happened: the AI couldn't do anything. No explanation, just a request for missing info.

- What still failed: everything. Zero usable output.

- What to try next: give it real context about the dataset.

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## Version 1 — Layer: Real context

**Prompt:** "Explain this data: a credit card customer dataset with ~9,000 rows and 18 features covering 6 months of behavior — balance, purchase frequency, credit limit, cash advance usage, payment history, etc."

**Output (excerpt):**

> This dataset captures customer credit card behavior over six months. Features like BALANCE and CREDIT_LIMIT describe account standing, PURCHASES and PURCHASE_FREQUENCY describe spending activity, and CASH_ADVANCE-related fields describe short-term borrowing behavior. Together, these suggest the data is suited for behavioral segmentation...

**Notes:**

- What changed in the prompt: added a real description of the dataset (size, timeframe, feature groups).

- What actually improved: the response went from "I need data" to a genuine explanation of what the columns represent and how they group together.

- What still failed: it's a dense, unstructured wall of text. No sense of who it's written for, and no mention of data-quality issues.

- What to try next: define an audience.

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## Version 2 — Layer: Defined audience

**Prompt:** *(V1) +* "Explain it to a non-technical manager who needs to understand what's possible before we start clustering — no jargon."

**Output (excerpt):**

> Think of this as a picture of how each customer uses their card — how much they spend, how often, and whether they rely on cash advances. Some customers pay off their balance responsibly; others carry debt or lean on short-term borrowing. Grouping customers by these patterns could reveal which ones are low-risk, high-value, or at risk of falling behind...

**Notes:**

- What changed in the prompt: added a specific audience and told it to drop jargon.

- What actually improved: technical terms disappeared (no more "categorical variable," "feature"), and the framing shifted from "what the columns are" to "what this means for the business."

- What still failed: still one long paragraph — nothing a busy manager could skim in 10 seconds. Also silently skips data-quality problems.

- What to try next: specify an output format.

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## Version 3 — Layer: Specified output format

**Prompt:** *(V2) +* "Format the answer as: (1) a one-paragraph summary, (2) a bullet list of what each feature group tells us, (3) a short 'watch out for' section on data-quality issues before clustering."

**Output (excerpt):**

> **Summary:** This data shows how each customer uses their card...

> **What each group tells us:**

> - Balance & limit → account standing

> - Purchases & frequency → spending activity

> - Cash advances → short-term borrowing reliance

> **Watch out for:** possible missing values, and a few features that may be skewed by a small number of heavy users.

**Notes:**

- What changed in the prompt: told it exactly how to structure the answer.

- What actually improved: went from a wall of text to something skimmable in seconds — the manager could scan the bullets and stop, or read the watch-out section if they care about data quality.

- What still failed: the "watch out for" section is generic boilerplate — it could be pasted onto any dataset. Nothing here is actually specific to *this* one.

- What to try next: add a constraint forcing specificity and honesty.

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## Version 4 — Layer: Constraints ⚠️ (this one backfired)

**Prompt:** *(V3) +* "Every claim about the data must name a specific feature. If you don't know an actual value, say so explicitly instead of guessing."

**Output (excerpt):**

> **Watch out for:** CASH_ADVANCE and PAYMENTS are typically right-skewed in datasets like this, though I can't confirm the actual skew without the real values. MINIMUM_PAYMENTS commonly has missing values in this kind of data, but again, I can't confirm the exact count without seeing it. I also can't confirm whether CREDIT_LIMIT has outliers without the real data...

**Notes:**

- What changed in the prompt: added a rule forcing it to name features and admit uncertainty instead of guessing.

- What actually improved: it stopped stating things with false confidence — assumptions are now flagged instead of hidden.

- **What still failed (the honest "made it worse" moment):** the answer got longer and repetitive — nearly a third of it became "I can't confirm this" disclaimers, repeated three separate times. For a busy manager, that's worse to read than Version 3, even though it's more honest.

- What to try next: add a review step to fix the honesty without the repetition.

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## Version 5 — Layer: Review instructions

**Prompt:** *(V4) +* "Before giving me the final answer, draft it, then review your own draft: collapse any repeated caveats into a single 'Assumptions' line at the end. Keep the rest confident and specific."

**Output (excerpt):**

> **Summary:** ...

> **What each group tells us:** ...

> **Watch out for:** CASH_ADVANCE and PAYMENTS are commonly right-skewed in this type of data; MINIMUM_PAYMENTS often has missing values worth checking before clustering.

> **Assumptions:** the skew and missing-value patterns above are typical for this dataset type, not confirmed against your actual file.

**Notes:**

- What changed in the prompt: added a self-review step that consolidates repeated caveats into one place.

- What actually improved: the hedging problem from V4 disappeared — the body reads clean and confident again, but the honesty is preserved in one clearly-labeled spot instead of scattered through the text.

- What still failed: it's still reasoning from general knowledge about "this type of dataset," not the actual numbers in the file — no prompt wording can fix that; it would need the real file attached.

- What to try next: attach the actual dataset — that's a different kind of change (data, not prompt), so the ladder stops here.

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## Final reusable prompt

> "Explain [dataset name/description] to a [specific audience] who needs [specific decision it informs], without jargon. Format the answer as: (1) a one-paragraph summary, (2) a bullet list of what each feature group tells us, (3) a 'watch out for' section on data-quality issues. Ground every claim in a named feature — if you don't know an actual value, say so rather than guessing. Draft the answer first, then review it yourself: collapse repeated caveats into a single 'Assumptions' line at the end so the rest stays confident and specific."

Anyone on the track can drop in their own dataset, audience, and decision, and reuse this as-is.